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Домашнее задание к занятию - Продвинутые методы работы с Terraform

Задание 1

1. Возьмите из [демонстрации к лекции готовый код](#) для создания с помощью двух вызовов remote-модуля -> двух VM, относящихся к разным проектам(marketing и analytics) используйте labels для обозначения принадлежности. В файле cloud-init.yml необходимо использовать переменную для ssh-ключа вместо хардкода. Передайте ssh-ключ в функцию template_file в блоке vars ={} . Воспользуйтесь [примером](#). Обратите внимание, что ssh-authorized-keys принимает в себя список, а не строку.
2. Добавьте в файл cloud-init.yml установку nginx.
3. Предоставьте скриншот подключения к консоли и вывод команды `sudo nginx -t`, скриншот консоли VM yandex cloud с их метками. Откройте terraform console и предоставьте скриншот содержимого модуля. Пример: > module.marketing_vm

Разбор решения задания 1

Прежде всего наполнил файлы с переменными своими значениями для токенов и id облака и каталога в Yandex Cloud. Соответствующие значения занес в файлы:

```
1 # файл providers.tf по аналогии с предыдущими заданиями
2 terraform {
3     required_providers {
4         yandex = {
5             source = "yandex-cloud/yandex"
6         }
7         template = {
8             source  = "hashicorp/template"
9             version = "~> 2.2"
10        }
11    }
12    required_version = ">1.12.0"
13 }
14
15 provider "yandex" {
16     cloud_id          = var.cloud_id
17     folder_id         = var.folder_id
18     zone              = var.default_zone
19     service_account_key_file = pathexpand("~/authorized_key.json")
20 }
21
```

```
22
23
24 # так же по аналогии с предыдущими заданиями задаю локально значение ssh ключа
    из файла на компьютере в файле main.tf
25
26
27 locals {
28     ssh_key = file("~/ssh/id_ed25519.pub")
29 }
30
31 # добавил значения в файл variables.tf
32 variable "cloud_id" {
33     type          = string
34     default       = "b1g3g3a6ro9rr9hd0mvm"
35     description = "https://cloud.yandex.ru/docs/resource-
manager/operations/cloud/get-id"
36 }
37
38 variable "folder_id" {
39     type          = string
40     default       = "b1grsjmidldjs2rmkgd0"
41     description = "https://cloud.yandex.ru/docs/resource-
manager/operations/folder/get-id"
42 }
```

```
43
44 variable "default_zone" {
45     type          = string
46     default       = "ru-central1-a"
47     description   = "https://cloud.yandex.ru/docs/overview/concepts/geo-scope"
48 }
49 variable "default_cidr" {
50     type          = list(string)
51     default       = ["10.0.1.0/24"]
52     description   = "https://cloud.yandex.ru/docs/vpc/operations/subnet-create"
53 }
54
55 variable "vpc_name" {
56     type          = string
57     default       = "develop"
58     description   = "VPC network&subnet name"
59 }
60
```

Создал файл `cloud-init.yml`

```
1 #cloud-config
2 users:
3     - name: ubuntu
```

```
4     groups: sudo
5     shell: /bin/bash
6     sudo: ["ALL=(ALL) NOPASSWD:ALL"]
7     ssh_authorized_keys:
8         - ${ssh_keys}
9
10    package_update: true
11    package_upgrade: false
12    packages:
13        - nano
14        - nginx
```

- В `packages` добавил `nano` вместо `vim` и `nginx` в соответствии с заданием, хотя это не имеет значения.

В файл `main.tf` добавил 2 вызова remote-модуля: `marketing_vm` и `analytics_vm`, и блок `templatefile`. В итоге получился файл:

```
1 resource "yandex_vpc_network" "develop" {
2     name = var.vpc_name
3 }
4
5 resource "yandex_vpc_subnet" "develop" {
6     name          = var.vpc_name
7     zone          = var.default_zone
```

```
8   network_id      = yandex_vpc_network.develop.id
9   v4_cidr_blocks  = var.default_cidr
10  }
11
12  locals {
13    ssh_key = file("~/.ssh/id_ed25519.pub")
14  }
15
16  data "template_file" "cloudinit" {
17    template = file("${path.module}/cloud-init.yml")
18    vars = {
19      ssh_keys = local.ssh_key
20    }
21  }
22
23  module "marketing_vm" {
24    source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
25    ref=main"
26    env_name       = "develop"
27    network_id     = yandex_vpc_network.develop.id
28    subnet_zones   = [var.default_zone]
29    subnet_ids     = [yandex_vpc_subnet.develop.id]
30    instance_name  = "marketing"
31    instance_count = 1
```

```
31 image_family    = "ubuntu-2004-lts"
32 public_ip       = true
33
34 labels = {
35     owner    = "team-marketing"
36     project  = "marketing"
37 }
38
39 metadata = {
40     user-data          = data.template_file.cloudinit.rendered
41     serial-port-enable = 1
42 }
43 }
44
45 module "analytics_vm" {
46     source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=main"
47     env_name        = "develop"
48     network_id       = yandex_vpc_network.develop.id
49     subnet_zones     = [var.default_zone]
50     subnet_ids       = [yandex_vpc_subnet.develop.id]
51     instance_name    = "analytics"
52     instance_count   = 1
53     image_family     = "ubuntu-2004-lts"
```

```
54     public_ip      = true
55
56     labels = {
57         owner      = "team-analytics"
58         project    = "analytics"
59     }
60
61     metadata = {
62         user-data      = data.template_file.cloudinit.rendered
63         serial-port-enable = 1
64     }
65 }
```

Для module `"marketing_vm"` и `module "analytics_vm"` использованы удаленные источники - модули по адресу: `source = "git:https://github.com/udjin10/yandex\_compute\_instance.git?ref=main"`

commands will detect it and remind you to do so if necessary.

```
• vboxuser@KVA:~/terraform/hw-04$ terraform init
```

Initializing modules...

Downloading git::https://github.com/udjin10/yandex_compute_instance.git?ref=main for analytics_vm...

- analytics_vm in .terraform/modules/analytics_vm

Downloading git::https://github.com/udjin10/yandex_compute_instance.git?ref=main for marketing_vm...

- marketing_vm in .terraform/modules/marketing_vm

Initializing provider plugins found in the configuration...

- Finding hashicorp/template versions matching "~> 2.2"...

- Reusing previous version of yandex-cloud/yandex from the dependency lock file

- Installing hashicorp/template v2.2.0...

- Installed hashicorp/template v2.2.0 (unauthenticated)

- Using previously-installed yandex-cloud/yandex v0.204.0

Initializing the backend...

Terraform has made some changes to the provider dependency selections recorded in the .terraform.lock.hcl file. Review those changes and commit them to your version control system if they represent changes you intended to make.

Warning: Incomplete lock file information for providers

Due to your customized provider installation methods, Terraform was forced to calculate lock file checksums locally for the following providers:

- hashicorp/template

The current .terraform.lock.hcl file only includes checksums for linux_amd64, so Terraform running on another platform will fail to install these providers.

To calculate additional checksums for another platform, run:

terraform providers lock -platform=linux amd64

(where linux_amd64 is the platform to generate)

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
• vboxuser@KVA:~/terraform/hw-04$
```

Консоль Yandex Cloud:

cloud-kost1n-valery default Compute Cloud / Виртуальные машины

Создать виртуальную машину

Виртуальные машины

Фильтр по имени или ID Все статусы Все платформы Все зоны доступности Требуется обслуживание

☐	Имя ↑↓	Статус ↑↓	ОС	Публичный IPv4	Идентификатор ↑↓	RAM	vCPU	Размер дисков	Метки	Cloud Backup	Платформа ↑↓	Доля vCPU	Прерываемая	Зона доступности ↑↓	Внутренний IPv4	Дата создания ↑↓	Группа размещения	Раздел (partition)	Пул резерво	
☐	develop-marketing-0	Running		51.250.12.218	fhm823bfm7m728ocnef1	1 ГБ	2	10 ГБ	owner : team-marketing +1	Не подключён	Intel Broadwell	5%	Да	ru-central1-a	10.0.1.30	24.05.2026, в 01:02	—	—	—	...
☐	develop-analytics-0	Running		62.84.125.216	fhuu4efnt8fmcjccv4k12	1 ГБ	2	10 ГБ	owner : team-analytics +1	Не подключён	Intel Broadwell	5%	Да	ru-central1-a	10.0.1.14	24.05.2026, в 01:02	—	—	—	...

cloud-kost1n-valery default Compute Cloud / Виртуальные машины / develop-marketing-0

develop-marketing-0
Виртуальная машина

Обзор

Диски и хранилища

Резервные копии

Права доступа

Операции

Мониторинг

Серийная консоль

Обзор

Идентификатор..... fhm823bfm7m728ocnef1

Статус..... Running

Имя..... develop-marketing-0

Дата создания..... 24.05.2026, в 01:02

Внутренний FQDN..... develop-marketing-0.ru-central1.internal

Зона доступности (?)..... ru-central1-a

Метки..... owner : team-marketing project : marketing

cloud-kost1n-valery default

Compute Cloud / Виртуальные машины / develop-analytics-0

develop-analytics-0
Виртуальная машина

Обзор

Диски и хранилища

Резервные копии

Права доступа

Операции

Мониторинг

Обзор

Идентификатор.....fhmu4efnt0fmjccv4k12

Статус.....Running

Имя.....develop-analytics-0

Дата создания.....24.05.2026, в 01:02

Внутренний FQDN.....develop-analytics-0.ru-central1.internal

Зона доступности (?).....ru-central1-a

Метки.....owner : team-analytics

project : analytics

Консоль Terraform

```
○ vboxuser@KVA:~/terraform/hw-04$ terraform console
> module.marketing_vm
{
  "all" = [
    {
      "allow_recreate" = tobool(null)
      "allow_stopping_for_update" = true
      "boot_disk" = tolist([
        {
          "auto_delete" = true
          "device_name" = "fhm9nvbvqna6vj6vlf15"
          "disk_id" = "fhm9nvbvqna6vj6vlf15"
          "initialize_params" = tolist([
            {
              "block_size" = 4096
              "description" = ""
              "image_id" = "fd8sjgpd03nsicrg3tlt"
              "kms_key_id" = ""
              "name" = ""
              "size" = 10
              "snapshot_id" = ""
              "type" = "network-hdd"
            },
          ])
          "mode" = "READ_WRITE"
        },
      ])
      "created_at" = "2026-05-23T21:05:38Z"
      "description" = "TODO: description; {{terraform yyy managed}}"
      "filesystem" = toset([])
      "folder_id" = "blgrsjmidldjs2rmkgd0"
      "fqdn" = "develop-marketing-0.ru-central1.internal"
      "gpu_cluster_id" = ""
      "hardware_generation" = tolist([
        {
          "generation2_features" = tolist([])
          "legacy_features" = tolist([
            {
              "pci_topology" = "PCI_TOPOLOGY_V2"
            },
          ])
        },
      ])
    },
  ],
}
```

```
  ])  
  "hostname" = "develop-marketing-0"  
  "id" = "fhmsrpj3ov8m5qiou"  
  "labels" = tomap({
```

```

    "owner" = "team-marketing"
    "project" = "marketing"
  })
  "local_disk" = tolist([])
  "maintenance_grace_period" = ""
  "maintenance_policy" = tostring(null)
  "metadata" = tomap({
    "serial-port-enable" = "1"
    "user-data" = <<-EOT
    users:
      - name: ubuntu
        groups: sudo
        shell: /bin/bash
        sudo: ["ALL=(ALL) NOPASSWD:ALL"]
        ssh_authorized_keys: ["ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICGCxz6Ttx4jaiMvpdANwL4va+6RTIWd+Vd8yLDNu1xD vboxuser@KVA\n"]

    package_update: true
    package_upgrade: false
    packages:
      - nano
      - nginx
    EOT
  })
  "metadata_options" = tolist([
    {
      "aws_v1_http_endpoint" = 1
      "aws_v1_http_token" = 2
      "aws_v2_http_endpoint" = 1
      "aws_v2_http_token" = 1
      "gce_http_endpoint" = 1
      "gce_http_token" = 1
    },
  ])
  "name" = "develop-marketing-0"
  "network_acceleration_type" = "standard"
  "network_interface" = tolist([
    {
      "dns_record" = tolist([])
      "index" = 0
      "ip_address" = "10.0.1.17"
      "ipv4" = true
      "ipv6" = false
      "ipv6_address" = ""
      "ipv6_dns_record" = tolist([])
    }
  ])

```

```

    "mac_address" = "d0:0d:1c:de:78:cc"
    "nat" = true
    "nat_dns_record" = tolist([])
    "nat_ip_address" = "93.77.184.148"
    "nat_ip_version" = "IPV4"
    "security_group_ids" = toset(null) /* of string */
    "subnet_id" = "e9b5v78mmrh2euc4paac"
  },
])
"placement_policy" = tolist([
  {
    "host_affinity_rules" = tolist([])
    "placement_group_id" = ""
    "placement_group_partition" = 0
  },
])
"platform_id" = "standard-v1"
"resources" = tolist([
  {
    "core_fraction" = 5
    "cores" = 2
    "gpus" = 0
    "memory" = 1
  },
])
"scheduling_policy" = tolist([
  {
    "preemptible" = true
  },
])
"secondary_disk" = toset([])
"service_account_id" = ""
"status" = "running"
"timeouts" = null /* object */
"zone" = "ru-central1-a"
},
]
"external_ip_address" = [
  "93.77.184.148",
]
"fqdn" = [
  "develop-marketing-0.ru-central1.internal",

```

```

]
"internal_ip_address" = [
  "10.0.1.17",
]
"labels" = [
  tomap({
    "owner" = "team-marketing"
    "project" = "marketing"
  }),
]
"network_interface" = [
  tolist([
    {
      "dns_record" = tolist([])
      "index" = 0
      "ip_address" = "10.0.1.17"
      "ipv4" = true
      "ipv6" = false
      "ipv6_address" = ""
      "ipv6_dns_record" = tolist([])
      "mac_address" = "d0:0d:1c:de:78:cc"
      "nat" = true
      "nat_dns_record" = tolist([])
      "nat_ip_address" = "93.77.184.148"
      "nat_ip_version" = "IPV4"
      "security_group_ids" = toset(null) /* of string */
      "subnet_id" = "e9b5v78mmrh2euc4paac"
    },
  ]),
]
}
>

```

Из консоли VM:

```

ubuntu@develop-marketing-0:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@develop-marketing-0:~$

```



```
ubuntu@develop-analytics-0:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@develop-analytics-0:~$
```

Задание 2

1. Напишите локальный модуль `vpc`, который будет создавать 2 ресурса: **одну** сеть и **одну** подсеть в зоне, объявленной при вызове модуля, например: `ru-central1-a`.
2. Вы должны передать в модуль переменные с названием сети, `zone` и `v4_cidr_blocks`.
3. Модуль должен возвращать в root module с помощью output информацию о `yandex_vpc_subnet`. Пришлите скриншот информации из terraform console о своем модуле. Пример: `> module.vpc_dev`
4. Замените ресурсы `yandex_vpc_network` и `yandex_vpc_subnet` созданным модулем. Не забудьте передать необходимые параметры сети из модуля `vpc` в модуль с виртуальной машиной.
5. Сгенерируйте документацию к модулю с помощью `terraform-docs`.

Пример вызова

```
1 module "vpc_dev" {
2   source      = "../vpc"
3   env_name    = "develop"
```

```
4     zone = "ru-central1-a"
5     cidr  = "10.0.1.0/24"
6 }
```

Разбор решения задачи 2

документация: <https://yandex.cloud/en/docs/terraform/concepts/modules>

Создаю локальный модуль `vpc` (в директории проекта папку `vpc`), в которой создаю следующие файлы

`vpc/variables.tf`

```
1  variable "env_name" {
2      type          = string
3      description = "Environment name"
4  }
5
6  variable "network_name" {
7      type          = string
8      description = "VPC network name"
9  }
10
11 variable "zone" {
12     type          = string
13     description = "Availability zone for subnet"
```

```
14 }
15
16 variable "v4_cidr_blocks" {
17     type          = list(string)
18     description = "IPv4 CIDR blocks for subnet"
19 }
```

`vpc/main.tf` Здесь `this` - имя ресурса внутри модуля

```
1 resource "yandex_vpc_network" "this" {
2     name = var.network_name
3 }
4
5 resource "yandex_vpc_subnet" "this" {
6     name          = "${var.env_name}-${var.zone}"
7     zone          = var.zone
8     network_id    = yandex_vpc_network.this.id
9     v4_cidr_blocks = var.v4_cidr_blocks
10 }
```

`vpc/outputs.tf`

```
1 output "network_id" {
2     description = "VPC network ID"
```

```
3     value          = yandex_vpc_network.this.id
4   }
5
6   output "subnet" {
7     description = "Subnet object"
8     value = {
9       id          = yandex_vpc_subnet.this.id
10      name         = yandex_vpc_subnet.this.name
11      zone         = yandex_vpc_subnet.this.zone
12      network_id   = yandex_vpc_subnet.this.network_id
13      v4_cidr_blocks = yandex_vpc_subnet.this.v4_cidr_blocks
14    }
15  }
```

vpc/providers.tf

```
1 terraform {
2   required_providers {
3     yandex = {
4       source = "yandex-cloud/yandex"
5     }
6   }
7 }
```

в основном файле `main.tf` в директории проекта добавил блок вместо ресурсов `yandex_vpc_network` и `yandex_vpc_subnet` на вызов локального модуля:

```
1 module "vpc_dev" {
2     source          = "./vpc"
3     env_name        = "develop"
4     network_name     = var.vpc_name
5     zone            = var.default_zone
6     v4_cidr_blocks   = var.default_cidr
7 }
8
9 locals {
10     ssh_key = file("~/.ssh/id_ed25519.pub")
11 }
12
13 data "template_file" "cloudinit" {
14     template = file("${path.module}/cloud-init.yml")
15     vars = {
16         ssh_keys = trimspace(local.ssh_key)
17     }
18 }
19
20 module "marketing_vm" {
```

```
21     source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=main"
22     env_name        = "develop"
23     network_id      = module.vpc_dev.network_id
24     subnet_zones    = [var.default_zone]
25     subnet_ids      = [module.vpc_dev.subnet.id]
26     instance_name   = "marketing"
27     instance_count  = 1
28     image_family    = "ubuntu-2004-lts"
29     public_ip       = true
30
31     labels = {
32         owner    = "team-marketing"
33         project = "marketing"
34     }
35
36     metadata = {
37         user-data          = data.template_file.cloudinit.rendered
38         serial-port-enable = 1
39     }
40 }
41
42 module "analytics_vm" {
```

```
43     source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=main"
44     env_name        = "develop"
45     network_id      = module.vpc_dev.network_id
46     subnet_zones    = [var.default_zone]
47     subnet_ids      = [module.vpc_dev.subnet.id]
48     instance_name    = "analytics"
49     instance_count   = 1
50     image_family     = "ubuntu-2004-lts"
51     public_ip        = true
52
53     labels = {
54         owner    = "team-analytics"
55         project  = "analytics"
56     }
57
58     metadata = {
59         user-data          = data.template_file.cloudinit.rendered
60         serial-port-enable = 1
61     }
62 }
```

ВМЕСТО `yandex_vpc_network.develop.id` теперь используется `module.vpc_dev.network_id` , а
ВМЕСТО `yandex_vpc_subnet.develop.id` - `module.vpc_dev.subnet.id` . Передача параметров

сети будет происходить из локального модуля в модули VM.

```
1 terraform init
2 terraform validate
3 terraform plan
```



```
• vboxuser@KVA:~/terraform/hw-04$ terraform plan
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 1s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
yandex_vpc_subnet.develop: Refreshing state... [id=e9b5v78mmrh2euc4paac]
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
yandex_vpc_network.develop: Refreshing state... [id=enpqqtshsrcs38hl6fpb2]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmu4efnt0fmjccv4k12]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm823bfm7m728ocnef1]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create
- ~ update in-place
- destroy

Terraform will perform the following actions:

```
# yandex_vpc_network.develop will be destroyed
# (because yandex_vpc_network.develop is not in configuration)
- resource "yandex_vpc_network" "develop" {
  - created_at      = "2026-05-23T21:05:35Z" -> null
  - default_security_group_id = "enplvj4bfc0r98vbg1tv" -> null
  - folder_id      = "b1grsjmidldjs2rmkgd0" -> null
  - id             = "enpqqtshsrcs38hl6fpb2" -> null
  - labels         = {} -> null
  - name           = "develop" -> null
  - subnet_ids     = [
    - "e9b5v78mmrh2euc4paac",
  ] -> null
  # (1 unchanged attribute hidden)
}
```

```
# yandex_vpc_subnet.develop will be destroyed
# (because yandex_vpc_subnet.develop is not in configuration)
- resource "yandex_vpc_subnet" "develop" {
  - created_at      = "2026-05-23T21:05:37Z" -> null
  - folder_id      = "b1grsjmidldjs2rmkgd0" -> null
  - id             = "e9b5v78mmrh2euc4paac" -> null
  - labels         = {} -> null
  - name           = "develop" -> null
  - network_id     = "enpqqtshsrcs38hl6fpb2" -> null
  - v4_cidr_blocks = [
    - "10.0.1.0/24",
  ] -> null
  - v6_cidr_blocks = [] -> null
  - zone           = "ru-central1-a" -> null
}
```

```

    # (2 unchanged attributes hidden)
  }

# module.analytics_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  id = "fhmu4efnt0fmjccv4k12"
  ~ metadata = {
    ~ "user-data" = <<-EOT
      #cloud-config
      users:
        - name: ubuntu
          groups: sudo
          shell: /bin/bash
          sudo: ["ALL=(ALL) NOPASSWD:ALL"]
          ssh_authorized_keys:
            - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICGCxz6Ttx4jaiMvpdANwL4va+6RTIwd+Vd8yLDNu1xD vboxuser@KVA

      package_update: true
      package_upgrade: false
      packages:
        - nano
        - nginx
    EOT
    # (1 unchanged element hidden)
  }
  name = "develop-analytics-0"
  # (15 unchanged attributes hidden)

  ~ network_interface {
    ~ subnet_id = "e9b5v78mmrh2euc4paac" -> (known after apply)
    # (10 unchanged attributes hidden)
  }

  # (5 unchanged blocks hidden)
}

# module.marketing_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  id = "fhm823bfm7m728ocnef1"
  ~ metadata = {
    ~ "user-data" = <<-EOT
      #cloud-config
      users:
        - name: ubuntu
          groups: sudo
          shell: /bin/bash

```

```

        sudo: ["ALL=(ALL) NOPASSWD:ALL"]
        ssh_authorized_keys:
          - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICGCxz6Ttx4jaiMvpdANwl4va+6RTIwd+Vd8ylDNu1xD vboxuser@KVA

      -
        package_update: true
        package_upgrade: false
        packages:
          - nano
          - nginx
      EOT
    # (1 unchanged element hidden)
  }
  name = "develop-marketing-0"
  # (15 unchanged attributes hidden)

~ network_interface {
  ~ subnet_id = "e9b5v78mmrh2euc4paac" -> (known after apply)
  # (10 unchanged attributes hidden)
}

# (5 unchanged blocks hidden)
}

# module.vpc_dev.yandex_vpc_network.this will be created
+ resource "yandex_vpc_network" "this" {
  + created_at      = (known after apply)
  + default_security_group_id = (known after apply)
  + folder_id       = (known after apply)
  + id              = (known after apply)
  + labels          = (known after apply)
  + name            = "develop"
  + subnet_ids      = (known after apply)
}

# module.vpc_dev.yandex_vpc_subnet.this will be created
+ resource "yandex_vpc_subnet" "this" {
  + created_at      = (known after apply)
  + folder_id       = (known after apply)
  + id              = (known after apply)
  + labels          = (known after apply)
  + name            = "develop-ru-central1-a"
  + network_id      = (known after apply)
  + v4_cidr_blocks = [
    + "10.0.1.0/24",
  ]
  + v6_cidr_blocks = (known after apply)
}

```

```
+ zone          = "ru-central1-a"
}

Plan: 2 to add, 2 to change, 2 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
vboxuser@KVA:~/terraform/hw-04$
```

Вывод консоли:

```
vboxuser@KVA:~/terraform/hw-04$ terraform console
> module.vpc_dev
{
  "network_id" = "enpqthsrcs38hl6fpb2"
  "subnet" = {
    "id" = "e9b5v78mmrh2euc4paac"
    "name" = "develop-ru-central1-a"
    "network_id" = "enpqthsrcs38hl6fpb2"
    "v4_cidr_blocks" = tolist([
      "10.0.1.0/24",
    ])
    "zone" = "ru-central1-a"
  }
}
> module.vpc_dev.subnet
{
  "id" = "e9b5v78mmrh2euc4paac"
  "name" = "develop-ru-central1-a"
  "network_id" = "enpqthsrcs38hl6fpb2"
  "v4_cidr_blocks" = tolist([
    "10.0.1.0/24",
  ])
  "zone" = "ru-central1-a"
}
>
```

Для работы с `terraform-docs` будет использована команда:

```
1 terraform-docs markdown table ./vpc > ./vpc/README.md
```

Прежде всего установил его с гитхаб репозитория (<https://terraform-docs.io/user-guide/installation/>)

Документация `terraform-docs`

<https://github.com/terraform-docs/terraform-docs>

<https://terraform-docs.io/user-guide/configuration/output/>

Результат terraform-docs:

Requirements

No requirements.

Provider

Name	Version
yandex	n/a

Modules

No modules.

Resources

Name	Type
yandex_vpc_network.this	resource
yandex_vpc_subnet.this	resource

Inputs

Name	Description	Type	Default	Required
env_name	Environment name	string	n/a	yes
network_name	VPC network name	string	n/a	yes
v4_cidr_blocks	IPv4 CIDR blocks for subnet	list(string)	n/a	yes
zone	Availability zone for subnet	string	n/a	yes

Outputs

Name	Description
network_id	VPC network ID
subnet	Subnet object

Запустил в гитхаб репозиторий

Задание 3

1. Выведите список ресурсов в стейте.
2. Полностью удалите из стейта модуль vpc.
3. Полностью удалите из стейта модуль vm.
4. Импортируйте всё обратно. Проверьте terraform plan. Значимых(!!) изменений быть не должно. Приложите список выполненных команд и скриншоты процессы.

Разбор решения задачи 3

Вывод списка ресурсов в стейте:

```
1 terraform state list
```

```
• vboxuser@KVA:~/terraform/hw-04$ terraform state list
data.template_file.cloudinit
module.analytics_vm.data.yandex_compute_image.my_image
module.analytics_vm.yandex_compute_instance.vm[0]
module.marketing_vm.data.yandex_compute_image.my_image
module.marketing_vm.yandex_compute_instance.vm[0]
module.vpc_dev.yandex_vpc_network.this
module.vpc_dev.yandex_vpc_subnet.this
○ vboxuser@KVA:~/terraform/hw-04$ █
```

Удаляю по очереди модуль vpc и модули vm.

```
• vboxuser@KVA:~/terraform/hw-04$ terraform state rm module.vpc_dev
Removed module.vpc_dev.yandex_vpc_network.this
Removed module.vpc_dev.yandex_vpc_subnet.this
Successfully removed 2 resource instance(s).
• vboxuser@KVA:~/terraform/hw-04$ terraform state list
data.template_file.cloudinit
module.analytics_vm.data.yandex_compute_image.my_image
module.analytics_vm.yandex_compute_instance.vm[0]
module.marketing_vm.data.yandex_compute_image.my_image
module.marketing_vm.yandex_compute_instance.vm[0]
• vboxuser@KVA:~/terraform/hw-04$ terraform state rm module.marketing_vm
Removed module.marketing_vm.data.yandex_compute_image.my_image
Removed module.marketing_vm.yandex_compute_instance.vm[0]
Successfully removed 2 resource instance(s).
• vboxuser@KVA:~/terraform/hw-04$ terraform state rm module.analytics_vm
Removed module.analytics_vm.data.yandex_compute_image.my_image
Removed module.analytics_vm.yandex_compute_instance.vm[0]
Successfully removed 2 resource instance(s).
• vboxuser@KVA:~/terraform/hw-04$ terraform state list
data.template_file.cloudinit
• vboxuser@KVA:~/terraform/hw-04$ █
```

Чтобы вернуть ресурсы обратно необходимо получить их из Yandex Cloud, поскольку они не удалены из облака физически, а только из стеита terraform

Для этого запрашиваю id ресурсов в `yc` (на выводе кроме актуальных еще есть остатки от предыдущих заданий)

```
1 yc vpc network list
2 yc vpc subnet list
3 yc compute instance list
```



```
vboxuser@KVA:~/terraform/hw-04$ yc vpc network list
```

ID	NAME
enp0sh2esfudgng27th4	kva
enpqqthsrcs38hl6fpb2	develop

There is a new yc version '1.9.0' available. Current version: '1.1.0'.
See release notes at <https://yandex.cloud/ru/docs/cli/release-notes>
You can install it by running the following command in your shell:
\$ yc components update

```
vboxuser@KVA:~/terraform/hw-04$ yc vpc subnet list
```

WARNING: Cannot connect to YC tool initialization service. Network connectivity to the service is required for cli version control. In case you are using yc in an isolated environment, you may turn off this warning by setting env YC_CLI_INITIALIZATION_SILENCE=true

ID	NAME	NETWORK ID	ROUTE TABLE ID	ZONE	RANGE
e2ldf4bq3ogc74gih7ki	default-ru-central1-b	enp0sh2esfudgng27th4		ru-central1-b	[10.129.0.0/24]
e9b0lut3pmjp16qqltrq	default-ru-central1-a	enp0sh2esfudgng27th4		ru-central1-a	[10.128.0.0/24]
e9b5v78mmrh2euc4paac	develop-ru-central1-a	enpqqthsrcs38hl6fpb2		ru-central1-a	[10.0.1.0/24]
fl81l4inrd5qn6l3732l	default-ru-central1-d	enp0sh2esfudgng27th4		ru-central1-d	[10.130.0.0/24]

```
vboxuser@KVA:~/terraform/hw-04$ yc compute instance list
```

WARNING: Cannot connect to YC tool initialization service. Network connectivity to the service is required for cli version control. In case you are using yc in an isolated environment, you may turn off this warning by setting env YC_CLI_INITIALIZATION_SILENCE=true

ID	NAME	ZONE ID	STATUS	EXTERNAL IP	INTERNAL IP
fhm823bfm7m728ocnef1	develop-marketing-0	ru-central1-a	RUNNING	51.250.12.218	10.0.1.30
fhm4efnt0fmjccv4k12	develop-analytics-0	ru-central1-a	RUNNING	62.84.125.216	10.0.1.14

Возвращаю все обратно через `import`

```
1 # Импорт сети
2 terraform import module.vpc_dev.yandex_vpc_network.this enpqqthsrcs38hl6fpb2
3
4 # Импорт подсети
```

```
5 terraform import module.vpc_dev.yandex_vpc_subnet.this e9b5v78mmrh2euc4paac
6
7 # Импорт BM marketing
8 terraform import 'module.marketing_vm.yandex_compute_instance.vm[0]'
  fhm823bfm7m728ocnef1
9
10 # Импорт BM analytics
11 terraform import 'module.analytics_vm.yandex_compute_instance.vm[0]'
  fhmu4efnt0fmjccv4k12
```

```
• vboxuser@KVA:~/terraform/hw-04$ terraform import module.vpc_dev.yandex_vpc_network.this enpqqtsrcs38hl6fpb2
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.vpc_dev.yandex_vpc_network.this: Importing from ID "enpqqtsrcs38hl6fpb2"...
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_network.this: Import prepared!
  Prepared yandex_vpc_network for import
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_network.this: Refreshing state... [id=enpqqtsrcs38hl6fpb2]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
```

Import successful!

The resources that were imported are shown above. These resources are now in your Terraform state and will henceforth be managed by Terraform.

Warning: Cannot connect to YC tool initialization service. Network connectivity to the service is required for provider version control.

```
with provider["registry.terraform.io/yandex-cloud/yandex"],
on /home/vboxuser/terraform/hw-04/providers.tf line 14, in provider "yandex":
14: provider "yandex" {
```

```
• vboxuser@KVA:~/terraform/hw-04$ terraform import module.vpc_dev.yandex_vpc_subnet.this e9b5v78mmrh2euc4paac
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_subnet.this: Importing from ID "e9b5v78mmrh2euc4paac"...
module.vpc_dev.yandex_vpc_subnet.this: Import prepared!
  Prepared yandex_vpc_subnet for import
module.vpc_dev.yandex_vpc_subnet.this: Refreshing state... [id=e9b5v78mmrh2euc4paac]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
```

Import successful!

The resources that were imported are shown above. These resources are now in your Terraform state and will henceforth be managed by Terraform.

Warning: Cannot connect to YC tool initialization service. Network connectivity to the service is required for provider version control.

```
with provider["registry.terraform.io/yandex-cloud/yandex"],
on /home/vboxuser/terraform/hw-04/providers.tf line 14, in provider "yandex":
14: provider "yandex" {
```

```
vboxuser@KVA:~/terraform/hw-04$ terraform import 'module.marketing_vm.yandex_compute_instance.vm[0]' fhm823bfm7m728ocnef1
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.yandex_compute_instance.vm[0]: Importing from ID "fhm823bfm7m728ocnef1"...
module.marketing_vm.yandex_compute_instance.vm[0]: Import prepared!
  Prepared yandex_compute_instance for import
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm823bfm7m728ocnef1]
```

Import successful!

The resources that were imported are shown above. These resources are now in your Terraform state and will henceforth be managed by Terraform.

Warning: Cannot connect to YC tool initialization service. Network connectivity to the service is required for provider version control.

```
with provider["registry.terraform.io/yandex-cloud/yandex"],
on /home/vboxuser/terraform/hw-04/providers.tf line 14, in provider "yandex":
14: provider "yandex" {
```

```
• vboxuser@KVA:~/terraform/hw-04$ terraform import 'module.analytics_vm.yandex_compute_instance.vm[0]' fhm4efnt0fmjccv4k12
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.analytics_vm.yandex_compute_instance.vm[0]: Importing from ID "fhmu4efnt0fmjccv4k12"...
module.analytics_vm.yandex_compute_instance.vm[0]: Import prepared!
  Prepared yandex_compute_instance for import
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm4efnt0fmjccv4k12]
```

Import successful!

The resources that were imported are shown above. These resources are now in your Terraform state and will henceforth be managed by Terraform.

Warning: Cannot connect to YC tool initialization service. Network connectivity to the service is required for provider version control.

```
with provider["registry.terraform.io/yandex-cloud/yandex"],
on /home/vboxuser/terraform/hw-04/providers.tf line 14, in provider "yandex":
14: provider "yandex" {
```

Проверяю что все корректно вернулось на место:

```
• vboxuser@KVA:~/terraform/hw-04$ terraform state list
data.template_file.cloudinit
module.analytics_vm.data.yandex_compute_image.my_image
module.analytics_vm.yandex_compute_instance.vm[0]
module.marketing_vm.data.yandex_compute_image.my_image
module.marketing_vm.yandex_compute_instance.vm[0]
module.vpc_dev.yandex_vpc_network.this
module.vpc_dev.yandex_vpc_subnet.this
```

В `terraform plan` незначительные изменения - `allow_stopping_for_update` - разрешения остановки для обновления

```

vboxuser@KVA:~/terraform/hw-04$ terraform plan
data.template_file.cloudinit: Reading...
data.template_file.cloudinit: Read complete after 0s [id=87b59fd41d0ffec9ac68bf912e62a84d3b6d1912254b94932a5dca6464c4565a]
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.vpc_dev.yandex_vpc_network.this: Refreshing state... [id=enpqqtsrcs38hl6fpb2]
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 1s [id=fd8sjgpd03nsicrg3tlt]
module.vpc_dev.yandex_vpc_subnet.this: Refreshing state... [id=e9b5v78mmrh2euc4paac]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm823bfm7m728ocnef1]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmu4efnt0fmjccv4k12]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# module.analytics_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  + allow_stopping_for_update = true
    id                        = "fhmu4efnt0fmjccv4k12"
    name                     = "develop-analytics-0"
    # (15 unchanged attributes hidden)

    # (6 unchanged blocks hidden)
}

# module.marketing_vm.yandex_compute_instance.vm[0] will be updated in-place
~ resource "yandex_compute_instance" "vm" {
  + allow_stopping_for_update = true
    id                        = "fhm823bfm7m728ocnef1"
    name                     = "develop-marketing-0"
    # (15 unchanged attributes hidden)

    # (6 unchanged blocks hidden)
}

Plan: 0 to add, 2 to change, 0 to destroy.

```

Ссылки на документацию: <https://developer.hashicorp.com/terraform/cli/commands/state/rm>
<https://developer.hashicorp.com/terraform/language/state/remove>
<https://developer.hashicorp.com/terraform/cli/import>
<https://developer.hashicorp.com/terraform/tutorials/state/state-import>

Дополнительные задания (со звёздочкой*)

Задание 4*

1. Измените модуль vpc так, чтобы он мог создать подсети во всех зонах доступности, переданных в переменной типа list(object) при вызове модуля.

Пример вызова

```
1 module "vpc_prod" {
2   source      = "./vpc"
3   env_name    = "production"
4   subnets = [
5     { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
6     { zone = "ru-central1-b", cidr = "10.0.2.0/24" },
7     { zone = "ru-central1-c", cidr = "10.0.3.0/24" },
8   ]
9 }
10
11 module "vpc_dev" {
12   source      = "./vpc"
13   env_name    = "develop"
14   subnets = [
15     { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
16   ]
17 }
```



```
17 }
```

Предоставьте код, план выполнения, результат из консоли YC.

Разбор решения задания 4*

Меняю содержимое файлов в папке vpc в директории проекта:

`vpc/variables.tf`

```
1  variable "env_name" {
2      description = "Environment name"
3      type        = string
4  }
5
6  variable "network_name" {
7      description = "VPC network name"
8      type        = string
9      default     = null
10 }
11
12 variable "subnets" {
13     description = "List of subnets to create"
14     type = list(object({
15         zone = string
16         cidr = string
```



```
17     })))  
18 }
```

`vpc/main.tf` добавляем `for_each`. Итоговое содержимое файла:

```
1 resource "yandex_vpc_network" "this" {  
2     name = var.network_name != null ? var.network_name : var.env_name  
3 }  
4  
5 resource "yandex_vpc_subnet" "this" {  
6     for_each = { for s in var.subnets : s.zone => s }  
7  
8     name          = "${var.env_name}-${each.value.zone}"  
9     zone          = each.value.zone  
10    network_id     = yandex_vpc_network.this.id  
11    v4_cidr_blocks = [each.value.cidr]  
12 }
```

В конструкции `name = var.network_name != null ? var.network_name : var.env_name` - если `network_name` передан то используем его (например `"production-network"`). Если не передан (= `null`) то используем `env_name` (`"develop"`). Это делает переменную `network_name` **необязательной**. **Тернарный оператор** - аналог `if/else`. Конструкция `for_each = { for s in var.subnets : s.zone => s } : for_each` требует на вход `map` (словарь), а тут `var.subnets` - это `list` поэтому `{ for s in var.subnets : s.zone`

=> `s` } превращает список в `map`, где ключ - зона (`ru-central1-a`). В данном случае `s` - временное значение или переменная цикла `each`.

Детально:

```
1 Было (list):
2 [
3   { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
4   { zone = "ru-central1-b", cidr = "10.0.2.0/24" },
5 ]
6
7 Стало (map):
8 {
9   "ru-central1-a" = { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
10  "ru-central1-b" = { zone = "ru-central1-b", cidr = "10.0.2.0/24" },
11 }
```

Ключ `map` (`s.zone`) становится **уникальным идентификатором** ресурса в state:

```
1 module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]
2 module.vpc_dev.yandex_vpc_subnet.this["ru-central1-b"]
```

Обращение к данным через `each`

```
1 name = "${var.env_name}-${each.value.zone}"
```

```
2 zone          = each.value.zone
3 v4_cidr_blocks = [each.value.cidr]
```

- `each.key` это ключ `map`, т.е. `"ru-central1-a"`
- `each.value` это весь объект `{ zone = "ru-central1-a", cidr = "10.0.1.0/24" }`
- `each.value.zone` и `each.value.cidr` это поля объекта

Terraform **запускает этот блок столько раз, сколько элементов в map**, то есть для каждой подсети отдельно.

`vpc/outputs.tf`

```
1 output "network_id" {
2   description = "VPC network ID"
3   value       = yandex_vpc_network.this.id
4 }
5
6 output "subnets" {
7   description = "Map of created subnets"
8   value       = yandex_vpc_subnet.this
9 }
```

Обновляем вызов модуля в корневом `main.tf`. Итоговое содержимое:

```
1 module "vpc_dev" {
```

```
2     source    = "./vpc"
3     env_name  = "develop"
4     subnets = [
5         { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
6     ]
7 }
8
9
10 module "vpc_prod" {                                # в итоге закоментил
11     source    = "./vpc"                             #
12     env_name  = "production"                         #
13     subnets = [                                     #
14         { zone = "ru-central1-a", cidr = "10.0.1.0/24" }, #
15         { zone = "ru-central1-b", cidr = "10.0.2.0/24" }, #
16         { zone = "ru-central1-d", cidr = "10.0.3.0/24" }, #
17     ]                                                 #
18 }                                                    #
19
20 locals {
21     ssh_key = file("~/ssh/id_ed25519.pub")
22 }
23
24 data "template_file" "cloudinit" {
25     template = file("${path.module}/cloud-init.yml")
```

```
26     vars = {
27         ssh_keys = trimspace(local.ssh_key)
28     }
29 }
30
31 module "marketing_vm" {
32     source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=main"
33     env_name        = "develop"
34     network_id      = module.vpc_dev.network_id
35     subnet_zones    = [var.default_zone]
36     subnet_ids      = [module.vpc_dev.subnets["ru-central1-a"].id]
37     instance_name   = "marketing"
38     instance_count  = 1
39     image_family     = "ubuntu-2004-lts"
40     public_ip       = true
41
42     labels = {
43         owner    = "team-marketing"
44         project  = "marketing"
45     }
46
47     metadata = {
48         user-data          = data.template_file.cloudinit.rendered
```

```
49     serial-port-enable = 1
50 }
51 }
52
53 module "analytics_vm" {
54     source          = "git::https://github.com/udjin10/yandex_compute_instance.git?
ref=main"
55     env_name        = "develop"
56     network_id      = module.vpc_dev.network_id
57     subnet_zones    = [var.default_zone]
58     subnet_ids      = [module.vpc_dev.subnet.id]
59     instance_name    = "analytics"
60     instance_count   = 1
61     image_family     = "ubuntu-2004-lts"
62     public_ip        = true
63
64     labels = {
65         owner    = "team-analytics"
66         project  = "analytics"
67     }
68
69     metadata = {
70         user-data          = data.template_file.cloudinit.rendered
71         serial-port-enable = 1
```

```
72     }
73 }
```

После всего этого:

```
1 terraform init
2 terraform validate
3 terraform plan
```

Для выполнения недостаточно квоты. Яндекс клауд дает создать только 2 сети. Одна создана по умолчанию, из создаваемых кодом в задании, зайдет только одна. Поэтому одну сеть закомментировал, вместо этого пример `for each` продемонстрировал на создании 2х подсетей. Отключил создание публичных ip на ВМ: `module "marketing_vm" ... public_ip = false` и `module "analytics_vm"` также из-за квоты.

В модуль `vpc_dev` добавил подсеть:

```
1 module "vpc_dev" {
2     source      = "../vpc"
3     env_name    = "develop"
4     subnets    = [
5         { zone = "ru-central1-a", cidr = "10.0.1.0/24" },
6         { zone = "ru-central1-b", cidr = "10.0.2.0/24" },
7     ]
8 }
```

8 }
9

В итоге создана сеть **develop** с двумя подсетями

cloud-kost1n-valery default Virtual Private Cloud / Облачные сети / develop

develop
Облачная сеть

Обзор

Таблицы маршрутизации

Группы безопасности

DNS зоны

Операции

Обзор

Идентификатор..... enpd67jump08fbrrr0f1

Статус..... Active

Имя..... develop

Дата создания..... 24.05.2026, в 23:03

Группа безопасности по умолчанию..... default-sg-enpd67jump08fbrrr0f1

Подсети

Имя ↑↓	Идентификатор ↑↓	Описание	IPv4 CIDR	IPv6 CIDR	Статус	Зона доступности	Метки	Таблица маршрутизации	
develop-ru-central1-b	e21bhtm7r0r861reaged	—	10.0.2.0/24	—	Active	ru-central1-b	—	—	...
develop-ru-central1-a	e9bj2vrutgge9m2o21fk	—	10.0.1.0/24	—	Active	ru-central1-a	—	—	...

А также 2 VM (надпись об ошибке это суточная квота на создание внешних ip, отключены в настройках TF)

cloud-kost1n-valery default Compute Cloud / Виртуальные машины

Создать виртуальную машину

Виртуальные машины

2 операции создания VM завершились с ошибкой за последние 10 минут. Подробная информация доступна в разделе [Операции](#).

Фильтр по имени или ID Все статусы Все платформы Все зоны доступности Требуется обслуживание

Имя ↑↓	Статус ↑↓	ОС	Публичный IPv4	Идентификатор ↑↓	RAM	vCPU	Размер дисков	Метки	Cloud Backup	Платформа ↑↓	Доля vCPU	Прерываемая	Зона доступности ↑↓	Внутренний IPv4	Дата создания ↑↓	Группа размещения	Раздел (partition)	Пул резервов	
develop-marketing-0	Running		—	fha7811bvs3cg9h8ctk6	1 ГБ	2	10 ГБ	owner: team-marketing +1	Не подключён	Intel Broadwell	5%	Да	ru-central1-a	10.0.1.9	24.05.2026, в 23:07	—	—	—	...
develop-analytics-0	Running		—	fhassjmeu1n5v7p7h2e1	1 ГБ	2	10 ГБ	owner: team-analytics +1	Не подключён	Intel Broadwell	5%	Да	ru-central1-a	10.0.1.32	24.05.2026, в 23:07	—	—	—	...

Скрин консоли с командой `terraform apply` :

```
vboxuser@KVA:~/terraform/hw-04$ terraform apply
module.vpc_dev.yandex_vpc_network.this: Refreshing state... [id=enpd67jump08fbrrr0fl]
module.analytics_vm.data.yandex_compute_image.my_image: Reading...
module.marketing_vm.data.yandex_compute_image.my_image: Reading...
module.analytics_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
module.marketing_vm.data.yandex_compute_image.my_image: Read complete after 0s [id=fd8sjgpd03nsicrg3tlt]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-a"]: Refreshing state... [id=e9bj2vrutgge9m2o21fk]
module.vpc_dev.yandex_vpc_subnet.this["ru-central1-b"]: Refreshing state... [id=e2lbhtm7r0r861reaged]
module.marketing_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhm3qeii1i134u14gfm6]
module.analytics_vm.yandex_compute_instance.vm[0]: Refreshing state... [id=fhmu615a9lmdrrqpehc7]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# module.analytics_vm.yandex_compute_instance.vm[0] will be created
+ resource "yandex_compute_instance" "vm" {
  + allow_stopping_for_update = true
  + created_at                 = (known after apply)
  + description                = "TODO: description; {{terraform yyy managed}}"
  + folder_id                  = (known after apply)
  + fqdn                       = (known after apply)
  + gpu_cluster_id             = (known after apply)
  + hardware_generation        = (known after apply)
  + hostname                   = "develop-analytics-0"
  + id                        = (known after apply)
  + labels                     = {
    + "owner"    = "team-analytics"
    + "project" = "analytics"
  }
  + maintenance_grace_period = (known after apply)
  + maintenance_policy       = (known after apply)
  + metadata                 = {
    + "serial-port-enable" = "1"
    + "user-data"          = <<-EOT
      #cloud-config
      users:
        - name: ubuntu
          groups: sudo
          shell: /bin/bash
          sudo: ["ALL=(ALL) NOPASSWD:ALL"]
          ssh_authorized_keys:
            - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICGCxz6Ttx4jaiMvpdANwL4va+6RTIWd+Vd8ylDNu1xD vboxuser@KVA
    + "user-data"          = <<-EOT
  }
}
```

```

        package_update: true
        package_upgrade: false
        packages:
          - nano
          - nginx
      EOT
    }
+ name = "develop-analytics-0"
+ network_acceleration_type = "standard"
+ platform_id = "standard-v1"
+ status = (known after apply)
+ zone = "ru-central1-a"

+ boot_disk {
  + auto_delete = true
  + device_name = (known after apply)
  + disk_id = (known after apply)
  + mode = (known after apply)

  + initialize_params {
    + block_size = (known after apply)
    + description = (known after apply)
    + image_id = "fd8sjgpd03nsicrg3tlt"
    + name = (known after apply)
    + size = 10
    + snapshot_id = (known after apply)
    + type = "network-hdd"
  }
}

+ metadata_options (known after apply)

+ network_interface {
  + index = (known after apply)
  + ip_address = (known after apply)
  + ipv4 = true
  + ipv6 = (known after apply)
  + ipv6_address = (known after apply)
  + mac_address = (known after apply)
  + nat = false
  + nat_ip_address = (known after apply)
}

```

```

    + subnet_id      = "e9bj2vrutgge9m2o21fk"
  }

+ placement_policy (known after apply)

+ resources {
  + core_fraction = 5
  + cores         = 2
  + memory        = 1
}

+ scheduling_policy {
  + preemptible = true
}
}

# module.marketing_vm.yandex_compute_instance.vm[0] will be created
+ resource "yandex_compute_instance" "vm" {
  + allow_stopping_for_update = true
  + created_at                = (known after apply)
  + description               = "TODO: description; {{terraform yyy managed}}"
  + folder_id                 = (known after apply)
  + fqdn                      = (known after apply)
  + gpu_cluster_id            = (known after apply)
  + hardware_generation       = (known after apply)
  + hostname                  = "develop-marketing-0"
  + id                        = (known after apply)
  + labels                    = {
    + "owner"   = "team-marketing"
    + "project" = "marketing"
  }
  + maintenance_grace_period = (known after apply)
  + maintenance_policy       = (known after apply)
  + metadata                  = {
    + "serial-port-enable" = "1"
    + "user-data"          = <<-EOT
      #cloud-config
      users:
        - name: ubuntu
          groups: sudo
          shell: /bin/bash
          sudo: ["ALL=(ALL) NOPASSWD:ALL"]
          ssh authorized keys:

```

```
- ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICGCxz6Ttx4jaiMvpdANw14va+6RTIWd+Vd8ylDNu1xD vboxuser@KVA
```

```
package_update: true
package_upgrade: false
packages:
  - nano
  - nginx
```

```
EOT
```

```
}
+ name = "develop-marketing-0"
+ network_acceleration_type = "standard"
+ platform_id = "standard-v1"
+ status = (known after apply)
+ zone = "ru-central1-a"

+ boot_disk {
  + auto_delete = true
  + device_name = (known after apply)
  + disk_id = (known after apply)
  + mode = (known after apply)

  + initialize_params {
    + block_size = (known after apply)
    + description = (known after apply)
    + image_id = "fd8sjgpd03nsicrg3tlt"
    + name = (known after apply)
    + size = 10
    + snapshot_id = (known after apply)
    + type = "network-hdd"
  }
}

+ metadata_options (known after apply)

+ network_interface {
  + index = (known after apply)
  + ip_address = (known after apply)
  + ipv4 = true
  + ipv6 = (known after apply)
  + ipv6_address = (known after apply)
  + mac_address = (known after apply)
```

```

+ nat          = false
+ nat_ip_address = (known after apply)
+ nat_ip_version = (known after apply)
+ subnet_id     = "e9bj2vrutgge9m2o21fk"
}

+ placement_policy (known after apply)

+ resources {
  + core_fraction = 5
  + cores         = 2
  + memory        = 1
}

+ scheduling_policy {
  + preemptible = true
}
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```

module.analytics_vm.yandex_compute_instance.vm[0]: Creating...
module.marketing_vm.yandex_compute_instance.vm[0]: Creating...
module.analytics_vm.yandex_compute_instance.vm[0]: Still creating... [00m10s elapsed]
module.marketing_vm.yandex_compute_instance.vm[0]: Still creating... [00m10s elapsed]
module.analytics_vm.yandex_compute_instance.vm[0]: Still creating... [00m20s elapsed]
module.marketing_vm.yandex_compute_instance.vm[0]: Still creating... [00m20s elapsed]
module.analytics_vm.yandex_compute_instance.vm[0]: Still creating... [00m30s elapsed]
module.marketing_vm.yandex_compute_instance.vm[0]: Still creating... [00m30s elapsed]
module.marketing_vm.yandex_compute_instance.vm[0]: Creation complete after 34s [id=fhm70llbvs3cg9h0ctk6]
module.analytics_vm.yandex_compute_instance.vm[0]: Still creating... [00m40s elapsed]
module.analytics_vm.yandex_compute_instance.vm[0]: Creation complete after 40s [id=fhmssjmeu1n5v7p7h2e1]

```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

vboxuser@KVA:~/terraform/hw-04\$

Ссылки на документацию: <https://developer.hashicorp.com/terraform/language/expressions/for>
https://developer.hashicorp.com/terraform/language/meta-arguments/for_each

<https://developer.hashicorp.com/terraform/language/expressions/conditionals>

<https://developer.hashicorp.com/terraform/language/values/variables>

Задание 6*

1. Используя готовый yandex cloud terraform module и пример его вызова(examples/simple-bucket): <https://github.com/terraform-yc-modules/terraform-yc-s3> . Создайте и не удаляйте для себя s3 бакет размером 1 ГБ(это бесплатно), он пригодится вам в ДЗ к 5 лекции.

Разбор решения задачи 6*

Решение задачи буду строить в отдельной папке внутри обвей директории модуля с решениями других задач: `~/terraform/hw-04/s3` . Модуль создания s3 бакета - **отдельный root module** со своим независимым `terraform.tfstate` изолированный от предыдущих задач. VPC и VM из прошлых задач или отдельно создаваемые не понадобятся.

```
1 mkdir -p ~/terraform/hw-04/s3
2 cd ~/terraform/hw-04/s3
```

Создаю файл `versions.tf`

```
1 terraform {
2     required_version = ">1.12.0"
3 }
```

```
4   required_providers {
5     yandex = {
6       source = "yandex-cloud/yandex"
7       version = "> 0.9"
8     }
9     aws = {
10      source = "hashicorp/aws"
11      version = "> 5.1"
12    }
13    random = {
14      source = "hashicorp/random"
15      version = "> 3.5"
16    }
17  }
18 }
```

Из модуля s3 из репозитория Yandex Cloud беру файл `providers.tf`

```
1 provider "yandex" {
2   cloud_id   = var.cloud_id
3   folder_id  = var.folder_id
4   zone       = var.default_zone
5   service_account_key_file = pathexpand("~/authorized_key.json")
6 }
```

```
7
8 provider "aws" {
9     region                = "us-east-1"
10    skip_credentials_validation = true
11    skip_requesting_account_id  = true
12    skip_metadata_api_check     = true
13    access_key                 = "mock_access_key"
14    secret_key                 = "mock_secret_key"
15 }
```

Создаю файл `variables.tf`

```
1
2 variable "cloud_id" {
3     type = string
4     default = "b1g3g3a6ro9rr9hd0mvm"
5     description = "[https://cloud.yandex.ru/docs/resource-
manager/operations/cloud/get-id](https://cloud.yandex.ru/docs/resource-
manager/operations/cloud/get-id%22)
6 }
7
8 variable "folder_id" {
9     type = string
10    default = "b1grsjmidldjs2rmkgd0"
```



```
11  description = "[https://cloud.yandex.ru/docs/resource-  
manager/operations/folder/get-id](https://cloud.yandex.ru/docs/resource-  
manager/operations/folder/get-id%22)  
12  }  
13  
14  variable "default_zone" {  
15  type = string  
16  default = "ru-central1-a"  
17  description = "[https://cloud.yandex.ru/docs/overview/concepts/geo-scope]"  
(https://cloud.yandex.ru/docs/overview/concepts/geo-scope%22)  
18  }
```

Создаю файл `main.tf`

```
1  module "s3" {  
2    source = "github.com/terraform-yc-modules/terraform-yc-s3"  
3  
4    bucket_name    = "netology-course-bucket-kva" # уникальное имя по всему облаку  
5    force_destroy  = false                        # флаг не удалять с объектами  
6    max_size       = 1073741824                   # размер 1 ГБ в байтах  
7  }
```

Создаю файл `outputs.tf`

```
1 output "bucket_name" {  
2     description = "S3 bucket name"  
3     value       = module.s3.bucket_name  
4 }
```

Запускаю

```
1 terraform init  
2 terraform validate  
3 terraform apply
```

```
vboxuser@KVA:~/terraform/hw-04/s3$ terraform apply_
module.s3.data.yandex_client_config.client: Read complete after 0s [id=3770268957]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# module.s3.random_string.unique_id will be created
+ resource "random_string" "unique_id" {
  + id          = (known after apply)
  + length      = 8
  + lower       = true
  + min_lower   = 0
  + min_numeric = 0
  + min_special = 0
  + min_upper   = 0
  + number      = true
  + numeric     = true
  + result      = (known after apply)
  + special     = false
  + upper       = false
}

# module.s3.yandex_storage_bucket.this will be created
+ resource "yandex_storage_bucket" "this" {
  + acl                  = (known after apply)
  + bucket               = "netology-course-bucket-kva"
  + bucket_domain_name   = (known after apply)
  + default_storage_class = "STANDARD"
  + folder_id            = "b1grsjmidldjs2rmkgd0"
  + force_destroy        = false
  + id                   = (known after apply)
  + max_size              = 1073741824
  + policy                = (known after apply)
  + website_domain        = (known after apply)
  + website_endpoint      = (known after apply)

  + anonymous_access_flags (known after apply)

  + grant (known after apply)
```

```

+ versioning (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ bucket_name = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

module.s3.random_string.unique_id: Creating...
module.s3.random_string.unique_id: Creation complete after 0s [id=8s1yj1or]
module.s3.yandex_storage_bucket.this: Creating...
module.s3.yandex_storage_bucket.this: Creation complete after 3s [id=netology-course-bucket-kva]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

bucket_name = "netology-course-bucket-kva"
vboxuser@KVA:~/terraform/hw-04/s3$

```

Бакет корректно создан

cloud-kost1n-valery
 default
 Object Storage / Бакеты

Бакеты

Имя ↑↓	Занято ↑↓	Кол-во объектов ↑↓	Доступ	
netology-course-bucket-kva	0 Б / 1 ГБ	0 объектов	С авторизацией	...

Пример бакета: <https://github.com/terraform-yc-modules/terraform-yc-s3/tree/master/examples/simple-bucket>

Документация: <https://yandex.cloud/ru/docs/storage/>

<https://yandex.cloud/ru/docs/storage/concepts/limits>

<https://yandex.cloud/ru/docs/storage/tf-ref>

https://registry.terraform.io/providers/yandex-cloud/yandex/latest/docs/resources/storage_bucket

Запустил результаты в гитхаб репозиторий (с ошибочным сообщением... hw-06 task 04 вместо hw-04 task 06...)

Задание 7*

1. Разверните у себя локально vault, используя docker-compose.yml в проекте.
2. Для входа в web-интерфейс и авторизации terraform в vault используйте токен "education".
3. Создайте новый секрет по пути <http://127.0.0.1:8200/ui/vault/secrets/secret/create> Path: example
secret data key: test secret data value: congrats!
4. Считайте этот секрет с помощью terraform и выведите его в output по примеру:

```
1 provider "vault" {  
2   address = "http://<IP_ADDRESS>:<PORT_NUMBER>"  
3   skip_tls_verify = true  
4   token = "education"
```

```
5 }
6 data "vault_generic_secret" "vault_example"{
7   path = "secret/example"
8 }
9
10 output "vault_example" {
11   value = "${nonsensitive(data.vault_generic_secret.vault_example.data)}"
12 }
13
14 Можно обратиться не к словарю, а конкретному ключу:
15 terraform console: >nonsensitive(data.vault_generic_secret.vault_example.data.<имя
ключа в секрете>)
```

5. Попробуйте самостоятельно разобраться в документации и записать новый секрет в vault с помощью terraform.

Разбор решения задания 7*

Создаю отдельную папку `vault` в директории домашнего задания:

```
1 mkdir -p ~/terraform/hw-04/vault
2 cd ~/terraform/hw-04/vault
```

Создаю `docker-compose.yml` :

```
1 version: "3"
2 services:
3   vault:
4     image: vault:1.13.3
5     container_name: vault
6     ports:
7       - "8200:8200"
8     environment:
9       VAULT_DEV_ROOT_TOKEN_ID: "education"
10      VAULT_DEV_LISTEN_ADDRESS: "0.0.0.0:8200"
11     cap_add:
12       - IPC_LOCK
```

`VAULT_DEV_ROOT_TOKEN_ID: "education"` - явно заданный токен от хранилища, только в учебной задаче

Запускаю:

```
1 docker compose up -d
2 docker ps                # проверяю что контейнер запущен
```

```
vboxuser@KVA:~/terraform/hw-04/vault$ docker compose up -d
WARN[0000] /home/vboxuser/terraform/hw-04/vault/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 8/8
✓ Image vault:1.13.3 Pulled 8.7s
✓ Network vault_default Created 0.1s
✓ Container vault Started 0.9s
vboxuser@KVA:~/terraform/hw-04/vault$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
529590ae19fd	vault:1.13.3	"docker-entrypoint.s..."	10 seconds ago	Up 9 seconds	0.0.0.0:8200->8200/tcp, [::]:8200->8200/tcp	vault

Создаю секрет через Web UI:

В браузере захожу на адрес: <http://127.0.0.1:8200>

Sign in to Vault

Method

Token

Token

Sign In

Contact your administrator for login credentials

1. Ввел **Token** → **education** → Sign In

http://127.0.0.1:8200/ui/vault/secrets

Secrets Engines

Enable new engine +

 cubbyhole/

cubbyhole_87a87b8c

per-token private secret storage

...

 secret/

v2 kv_081a5000

key/value secret storage

...

2. Перешел по ссылке из задания: <http://127.0.0.1:8200/ui/vault/secrets/secret/create>

3. Заполнил:

- **Path:** **example**
- **Key:** **test**
- **Value:** **congrats!**

4. Сохранил: **Save**

 http://127.0.0.1:8200/ui/vault/secrets/secret/show/example

[secret](#) [example](#)

example

[Secret](#) [Metadata](#)

 JSON

Delete

Copy ▾

Version 1 ▾

Create new version +

Key

Value

Version created May 25, 2026 06:00 PM

test

  congrats!

Создал папку для Terraform:

```
1 mkdir -p ~/terraform/hw-04/vault/terraform
2 cd ~/terraform/hw-04/vault/terraform
```

В этой директории:

Создал файл `versions.tf`:

```
1 terraform {
2   required_providers {
3     vault = {
4       source  = "hashicorp/vault"
5       version = "~> 3.0"
6     }
7   }
8 }
```

Создал файл `main.tf`

```
1 provider "vault" {
2   address          = "http://127.0.0.1:8200"
3   skip_tls_verify  = true
4   token            = "education"
5 }
```

```
6
7 data "vault_generic_secret" "vault_example" {
8     path = "secret/example"
9 }
10
11 resource "vault_generic_secret" "my_secret" {
12     path      = "secret/my_tf_secret"
13     data_json = jsonencode({
14         username = "admin"
15         password = "superpassword123"
16     })
17 }
```

Здесь: `data "vault_generic_secret" "vault_example"` читает секрет по пути `path = "secret/example"`

Блок `resource "vault_generic_secret" "my_secret"` записывает результат в формат json

Создал файл `outputs.tf`

```
1
2 output "vault_example" {
3     value = nonsensitive(data.vault_generic_secret.vault_example.data)
4 }
5
```

```
6
7 output "vault_example_test_key" {
8     value = nonsensitive(data.vault_generic_secret.vault_example.data["test"])
9 }
```

`value = nonsensitive` явно указанное значение чтобы terraform не пометил в output ключи и секреты как sensitive данные, как это всегда происходит по умолчанию.

Запустил из субдиректории `~/terraform/hw-04/vault/terraform$`:

```
1 terraform init
2 terraform validate
3 terraform plan
4 terraform apply
```

terraform console:

```
1 terraform console
2
3 > nonsensitive(data.vault_generic_secret.vault_example.data)
4 > nonsensitive(data.vault_generic_secret.vault_example.data["test"])
```

```
• vboxuser@KVA:~/terraform/hw-04/vault/terraform$ terraform apply
data.vault_generic_secret.vault_example: Reading...
data.vault_generic_secret.vault_example: Read complete after 0s [id=secret/example]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# vault_generic_secret.my_secret will be created
+ resource "vault_generic_secret" "my_secret" {
  + data           = (sensitive value)
  + data_json      = (sensitive value)
  + delete_all_versions = false
  + disable_read    = false
  + id             = (known after apply)
  + path           = "secret/my_tf_secret"
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ vault_example = {
  + test = "congrats!"
}
+ vault_example_test_key = "congrats!"
```

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
vault_generic_secret.my_secret: Creating...
vault_generic_secret.my_secret: Creation complete after 0s [id=secret/my_tf_secret]
```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
vault_example = tomap({
  "test" = "congrats!"
})
vault_example_test_key = "congrats!"
```

```
> nonsensitive(data.vault_generic_secret.vault_example.data)
tomap({
  "test" = "congrats!"
})
> nonsensitive(data.vault_generic_secret.vault_example.data["test"])
"congrats!"
> █
```

Запустил изменения в гитхаб репозиторий

Документация **Vault + Docker**:

<https://hub.docker.com/r/hashicorp/vault>

<https://developer.hashicorp.com/vault/docs/concepts/dev-server>

Terraform провайдер Vault:

- Обзор: registry.terraform.io/providers/hashicorp/vault/latest/docs
- Настройка: registry.terraform.io/providers/hashicorp/vault/latest/docs#provider-arguments
- Чтение секрета: registry.terraform.io/providers/hashicorp/vault/latest/docs/data-sources/generic_secret
- Запись секрета: registry.terraform.io/providers/hashicorp/vault/latest/docs/resources/generic_secret
- **Функция** `nonsensitive()` :
developer.hashicorp.com/terraform/language/functions/nonsensitive]
(<https://developer.hashicorp.com/terraform/language/functions/nonsensitive>)
- Концепция sensitive values в Terraform: developer.hashicorp.com/terraform/language/values/outputs#sensitive-suppressing-

[values-in-cli-output](#)

KV Secrets Engine (тип хранилища):

- KV v1: developer.hashicorp.com/vault/docs/secrets/kv/kv-v1
- KV v2: developer.hashicorp.com/vault/docs/secrets/kv/kv-v2